Script-Controlled ACL – Restrict Record Access Based on Field Value

---

Team Information

Team ID: SWTID-2026-9693

Team Size: 4

Team Leader

Name: GIRI G

Email: ganeshangiri08@gmail.com

Team Members

Name: SHAND K

Email: shanthk296@gmail.com

Name: DEVAA R

Email: rdevaa500@gmail.com

Name: ABISHEK S

Email: abishekkabaddii07@gmail.com

Introduction

In modern applications, securing sensitive data is critical. Traditional role-based access may not be sufficient when access needs to be controlled dynamically. This project introduces a script-based ACL mechanism where access is granted or denied based on field-level conditions within records

Description:

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

Creation of Users and Roles

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
   - Navigate to **User Administration → Users**
   - Click **New**
   - **User ID**: EEE User

- **First name**: EEE
- **Last name**: User
- **Email**: eeeuser@gmail.com
- Save the record

Role Creation

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

Tables Creation

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
   - **Label:** Institution Details
   - **Name:** u_institution_details
   - **Extends for:** false
3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
   - **Student Roll Number**– Auto Number
   - **Student Name** – Reference- User
   - **Faculty Name** – Reference- User
   - **Branch** – Choice (ECE, EEE, CSE)
   - **Email**– String
   - **Phone Number**– String
   - **Description**– Multi String
5. Save the Table
6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

Creation of Access Control Lists(Read ACL)

1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
   - **Type:** record
   - **Operation:** read
   - **Name:** u_institution_details
   - **Active:** true
   - **Advanced:** true
5. In the **Requires role** related list, add:
   - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {
  // Allow admin users full access
  if (gs.hasRole('admin')) {
    return true;
  }
  // Allow only EEE branch users to see EEE records
  if (gs.hasRole('bb1')){
    return true;
  }
  // Deny access for all others
  return false;
})();
```

8. Click **Submit** to save the ACL.
9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:

   - **Type:** record
   - **Operation:** create
   - **Name:** u_institution_details
   - **Active:** true

3. In the **Requires role** related list, add:

   - bb2 (custom role)

4. No need to add any Data condition
5. Save the ACL Record

Creation of Access Control Lists(Write ACL)

1. Click New to create a new ACL(Write)
2. Enter the following details:

   - Type: record
   - Operation: Write
   - Name: u_institution_details
   - Active: true

3. In the Requires role related list, add:

   - bb3 (custom role)

4. No need to add any Data condition
5. Save the ACL Record

Creation of Access Control Lists(Delete ACL)

1. Click **New** to create a new ACL(Delete)
2. Enter the following details:

   - **Type:** record
   - **Operation:** Delete
   - **Name:** u_institution_details
   - **Active:** true

3. In the **Requires role** related list, add:

   - bb4 (custom role)

4. No need to add any Data condition
5. Save the ACL Record

Outcome:

By completing this project, learners gain a clear understanding of:
- How Script-Controlled ACLs work in ServiceNow
- Implementation of READ, WRITE, CREATE, DELETE access controls
- How field values influence record access decisions
- How ACLs protect sensitive data from unauthorized actions
- Application across forms, lists, and Service Portal

Future Enhancements:

- Integration with AI-based security systems
- Advanced user behavior tracking
- Real-time alerts for unauthorized access attempts

References:

- Official Documentation (e.g., ServiceNow ACL Docs)
- Online tutorials and research papers

Conclusion

This project provides practical knowledge of implementing secure access control mechanisms in ServiceNow using ACLs. It ensures that only authorized users can interact with specific data based on predefined conditions